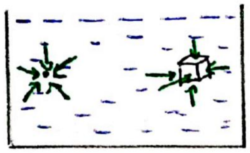
Fluid Mechanics-1

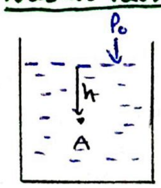
Pressure inside a fluid body :

→ Inside a fluid, pressure always acts in all dirⁿs on a point.

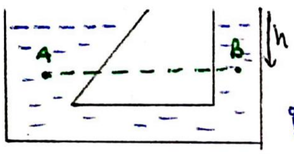
→ On a body, pressure acts ⊥ar to its surface in the fluid.



Pressure Variation :→



$$\boxed{P_A = P_0 + \rho g h}$$



(i) Pressure at same level is same, across same liq.

i.e $\boxed{P_A = P_B = P_0 + \rho g h}$

(ii) Pressure at h-depth is independent of shape of vessel.

(iii) $\boxed{P_0 = 1 \text{ atm} = 1.013 \times 10^5 \text{ Pa} \simeq 10^5 \text{ Pa}}$

Specific Gravity / Relative Density :→

For any substance,

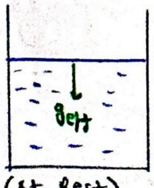
$$\boxed{\rho_s = \frac{\rho}{\rho_w}} \text{ @ } 4°C$$

ρ → Density of substance
ρ_w → Density of water

Free Surface :

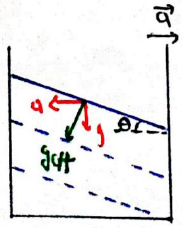
→ Free surface of a fluid is always ⊥ar to g_{eff} of container.

a)



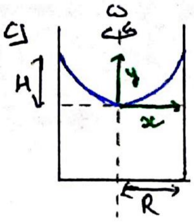
(At Rest)

b)



$\boxed{\tan\theta = a/g}$

c)



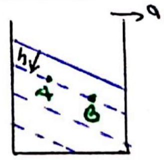
Here, $\boxed{y = \frac{\omega^2 x^2}{2g}}$

And, if $x = R$ & $y = H$

$$\Rightarrow \boxed{H = \frac{\omega^2 R^2}{2g}}$$

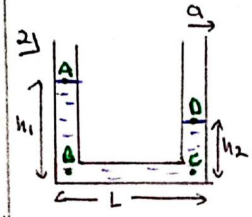
Pressure Variation in Accelerated System :

1)



$$\boxed{P_A = P_B = P_0 + \rho \cdot g_{eff} \cdot h}$$

2)



→ Here, diameter of tube is very small.

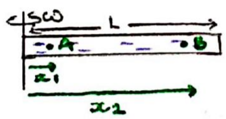
→ Pressure is taken -ve, in opp. dirⁿ of acceleration.

So, from point A to D

$$P_A + \rho g h_1 - \rho a L - \rho g h_2 = P_D$$

$$\Rightarrow \boxed{h_1 - h_2 = \frac{aL}{g}} \quad (\because P_A = P_D = P_0)$$

3)



$$\boxed{P_B - P_A = \frac{1}{2}\rho\omega^2 (x_2^2 - x_1^2)}$$

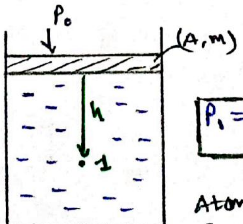
↳ If $x = 0$ at axis, then

$$\boxed{\Delta P = \frac{1}{2}\rho\omega^2 x_2^2}$$

Pascals's law :

→ External pressure gets evenly distributed in all directions.

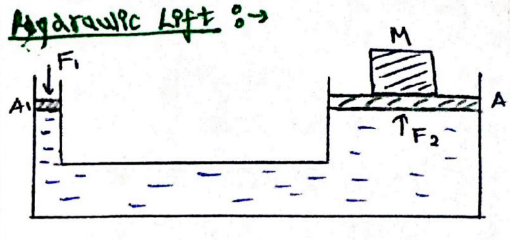
Eg :



$$\boxed{P_1 = P_0 + \frac{Mg}{A} + \rho g h}$$

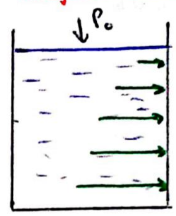
Atmospheric pressure & pressure due to piston gets distributed to every point of the fluid evenly.

Hydraulic Lift :→



Here, $\boxed{F_2 = \left(\frac{F_1}{A_1}\right) A_2}$ $\quad \because A_2 >>> A_1 \Rightarrow F_2 > F_1$

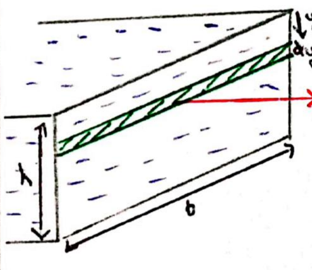
Avg Pressure on side walls :



→ The pressure on side walls keep increasing with the depth of fluid.

$$\boxed{P_{avg} = \frac{\rho g h}{2}}$$

Force & Torque on side walls :



$dF = P(y) \times A$

$dF = \rho g y \times b\,dy$

$\Rightarrow F = \rho g b \int_0^h y \cdot dy$

$\Rightarrow \boxed{F = \frac{\rho g h}{2} \times bh}$ ← ($P_{avg} \times$ Area of wall)

→ For torque

$d\tau = dF \times (h - y)$

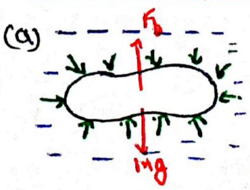
$d\tau = \rho g y \cdot b\,dy\,(h-y)$

$\Rightarrow \tau = \rho g b \int_0^h y(h-y)\,dy$

$\Rightarrow \boxed{\tau = \frac{\rho g b h^3}{6}}$ and $\boxed{y = \frac{2h}{3}}$

Archimedes Principle / Floatation :

(a)



→ Buoyant Force,

$$\boxed{F_b = \rho V g}$$

(V → Vol. displaced)

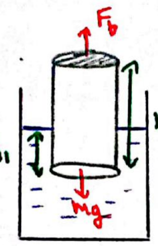
→ If body is solid, then
(i) Body Sinks ⇒ $\rho_s > \rho_L$
(ii) Body Floats ⇒ $\rho_s \leq \rho_L$

Body Floats ($\rho_s \le \rho_L$)

Some portion of body is outside ($\rho_s < \rho_L$)

Body completely submerged ($\rho_s = \rho_L$)

(b)



(i) At equilibrium

$$F_b = mg$$

$$\rho_L V_1 g = \rho_s V_0 g$$

$$\Rightarrow \boxed{\frac{V_1}{V_0} = \frac{\rho_s}{\rho_L}}$$

[$V_1 \to$ Vol. displaced by solid
$V_0 \to$ Vol. of solid]

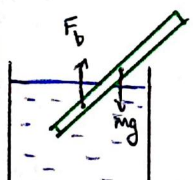
(ii) Since, $\rho_L V g = \rho_s V_0 g$

$$\rho_L (Ah_1) g = \rho_s (Ah) g$$

$$\Rightarrow \boxed{\frac{h_1}{h_0} = \frac{\rho_s}{\rho_L}}$$

[$h_1 \to$ Heigh sinked
$h \to$ Complete height]

Centre of Buoyancy :



→ C.O.B is C.O.M of the part submerged in fluid.

→ F_b acts through C.O.B, just like mg acts through C.O.M.

Apparent weight:

↳ It is the decreased weight of a body, when it is submerged in a fluid, due to buoyant Force (F_b).

$$W = mg - F_b$$

$$\Rightarrow \boxed{W = mg\left(1 - \frac{\rho_L}{\rho_s}\right)}$$

Reading on weighing scale

$$\boxed{\text{Reading} = m\left(1 - \frac{\rho_L}{\rho_s}\right)}$$

(Fluid Dynamics Start ⟶)

Reynold's No. :

$$\boxed{Re = \frac{\rho v d}{\eta}}$$ (Dimensionless quantity)

$\rho \to$ Density of liq.

$\eta \to$ Coeff of viscosity

$v \to$ Velocity of fluid

$d \to$ Diameter of cross section

If. $Re < 10^3 \longrightarrow$ Streamline flow

$Re > 2\times10^3 \to$ Turbulent flow

$10^3 < Re < 2\times10^3 \to$ Unsteady flow

→ Laws of fluid dynamics are applicable only on ideal fluid having streamline flow.

Ideal Fluid

- → Incompressible
- → Non-viscous
- → Streamline flow
- → No Rotation Effect.

Critical Velocity (v_c) :

→ It is the velocity of fluid, above which its flow becomes turbulent from streamline.

$$\boxed{v_c = Re\left(\frac{\eta}{\rho d}\right)}$$

($d \to$ Diameter of cross section)

$v_c \propto \eta$ ¦ $v_c \propto \frac{1}{\rho}$ ¦ $v_c \propto \frac{1}{d}$ or $\frac{1}{r}$

Equation of Continuity :

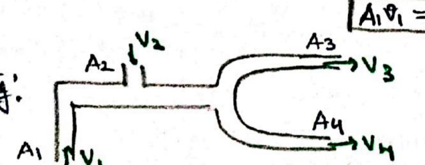
→ Volume flow rate of a fluid is constant

i.e $\frac{dV}{dt} = A\cdot\frac{dx}{dt} \Rightarrow \boxed{A\cdot v = \text{const}}$

$$\Downarrow$$

$$\boxed{A_1 v_1 = A_2 v_2}$$

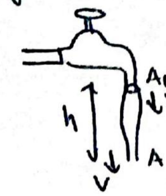
Eg:



$$(\Sigma Av)_{\text{Entering}} = (\Sigma Av)_{\text{Exiting}}$$

$$A_1v_1 + A_2v_2 = A_3v_3 + A_4v_4$$

Eg: Find the area A, after heigh h.



Sol^n Since, $A_0V_0 = AV$

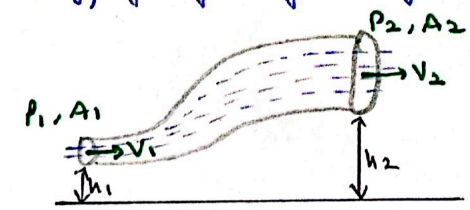
$\therefore v^2 = u^2 + 2as$

$v^2 = V_0^2 + 2gh$

$$\Rightarrow \boxed{A = \frac{A_0V_0}{\sqrt{V_0^2 + 2gh}}}$$

Bernoulli's Equation :

→ Based on conservation of mechanical energy for flow of ideal fluid.



$$\boxed{P + \frac{1}{2}\rho v^2 + \rho gh = \text{const.}}$$

$P \to$ Pressure Energy / vol.

$\frac{1}{2}\rho v^2 \to$ Kinetic Energy / vol.

$\rho gh \to$ Potential Energy / vol.

For above diagram

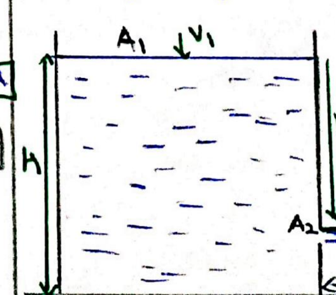
$$\left(P_1 + \frac{1}{2}\rho v_1^2 + \rho gh_1\right) = \left(P_2 + \frac{1}{2}\rho v_2^2 + \rho gh_2\right)$$

→ For horizontal flow

$$\boxed{v \propto \frac{1}{P}}$$ If $V_1 > V_2$, $P_1 < P_2$

Torricelli's Law :

→ Speed of efflux from an open tank is identical to a freshly falling body.



$A_1 >>> A_2$

$\Rightarrow V_1 <<< V_2$

(Negligible)

(i) Speed of efflux

$$\boxed{V_2 = \sqrt{2gh}}$$

(ii) Range of efflux

$$\boxed{R = V_2 \times \sqrt{\frac{2(H-h)}{g}}} \quad \left(\because t = \sqrt{\frac{2h}{g}}\right)$$

Free fall

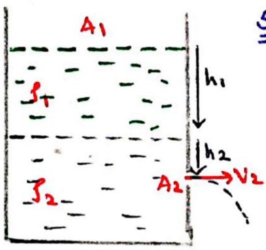
(iii) For Range to be max

$\rightarrow \frac{dR}{dh} = 0$

$\rightarrow \boxed{h = H/2}$ (Hole should be at mid-point)

$\rightarrow \boxed{R_{max} = H}$

Eg:



Solⁿ By energy conservation

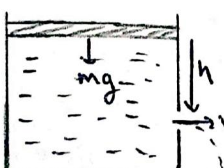
$\begin{pmatrix}\text{Loss in} \\ P\end{pmatrix} = \begin{pmatrix}\text{Gain} \\ \text{in KE}\end{pmatrix}$

$\Rightarrow (P_0 + \rho_1 g h_1 + \rho_2 g h_2) - P_0 = \frac{1}{2}\rho_2 V_2^2$

($V_1 \rightarrow$ Neglected)

$$\Rightarrow \boxed{\rho_1 g h_1 + \rho_2 g h_2 = \frac{1}{2}\rho_2 V_2^2}$$

Eg:



Solⁿ By energy conservation

$\left(P_0 + \rho g h + \frac{mg}{A}\right) - P_0 = \frac{1}{2}\rho V_2^2$

$$\Rightarrow \boxed{\rho g h + \frac{mg}{A} = \frac{1}{2}\rho V_2^2}$$

Thrust on Vessel :-

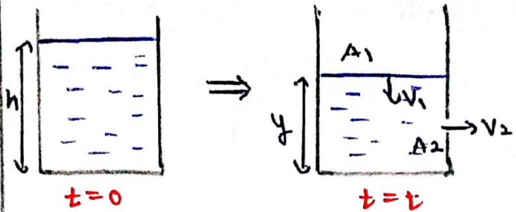
When the efflux exits vessel, it applies a backward thrust on vessel. (Newton's 3rd law)

$$\boxed{F_{Th} = \rho A V^2 = \rho A (2gh)}$$

$A \rightarrow$ Area of hole

$\rho \rightarrow$ Density of efflux fluid.

Vessel Emptying Time :-



(i) Time to empty vessel when it is completely filled

$$\boxed{t = \frac{A_1}{A_2}\sqrt{\frac{2H}{g}}}$$

(ii) Time to empty a partially filled vessel, which has been already leaked.

$$\boxed{t = \frac{A_1}{A_2}\sqrt{\frac{2}{g}}\left(\sqrt{H} - \sqrt{y}\right)}$$

Venturimeter :

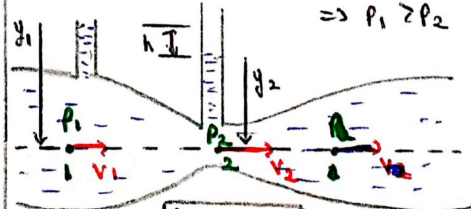
$\rightarrow$ It is a device used to measure speed of flow of fluid.

(a) Setup-1 :

$A_1 > A_2$

$\Rightarrow V_1 < V_2$

$\Rightarrow P_1 > P_2$

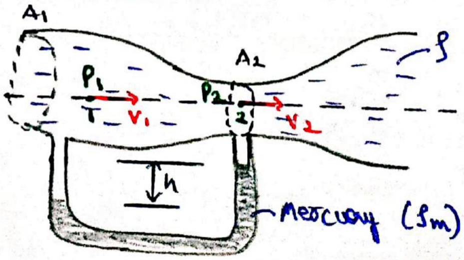


$$\boxed{V_2 = \frac{A_1 V_1}{A_2}}$$

For above setup

$$\boxed{V_1 = \sqrt{\frac{2gh}{\left(\frac{A_1}{A_2}\right)^2 - 1}}}$$

(b) Setup-2 :



Here,

$$\boxed{V_1 = \sqrt{\frac{2\rho_m g h}{\rho\left[\left(\frac{A_1}{A_2}\right)^2 - 1\right]}}}$$

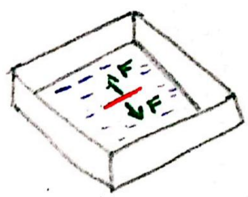
Fluid Properties

Surface Tension :

↳ Property of liquid where its surface

(i) tries to have minimum area

(ii) Acts as a stretched membrane.



Surface Tension,

$$\boxed{S = \frac{F}{L}}$$

$$\Rightarrow \boxed{F = SL}$$

$\rightarrow$ F acts ⊥ar to line and tangential to free surface.

Lifting of Bodies :

a) $F_{ext} = mg + 2 \cdot S(\pi r 2)$ — circumference

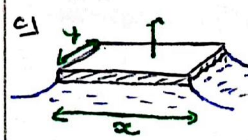
Because two surfaces (inner & outer)

(Ring)

b) $F_{ext} = mg + S.(2\pi r)$

(Coin)

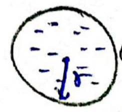
c) $F_{ext} = mg + S(2x + 2y)$



(Rectangular Lamina)

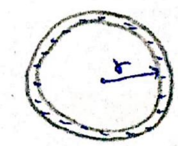
Surface Energy :

$$\boxed{S = \frac{E}{A}} \Rightarrow \boxed{E = S.A}$$



(Drop)

$E = S.4\pi r^2$



(Bubble)

$E = 2 \times (S.4\pi r^2)$

Formation of larger Drop :

o o o o - - - o ⟶ O

(n-drops) r

→ Since volume remains constant

$$\frac{4}{3}\pi R^3 = n\cdot\frac{4}{3}\pi r^3$$

$$\Rightarrow \boxed{R = r\cdot n^{1/3}}$$

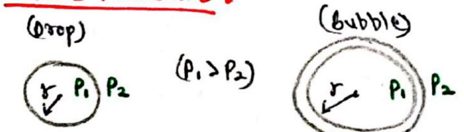
Now, here Area ↓ ⇒ Energy ↓

$E_i = n\cdot S\cdot 4\pi r^2 \qquad E_f = S\cdot 4\pi R^2 = S\cdot 4\pi r^2\cdot n^{2/3}$

→ $E_i - E_f$ increases temp. of bigger drop by $\Delta\theta$.

$$\boxed{\Delta E = E_i - E_f = m\cdot s\cdot \Delta\theta}$$

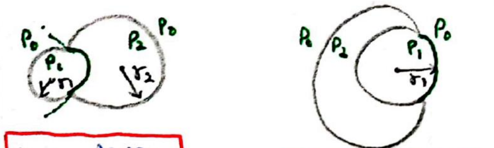
Excess Pressure :



$$\boxed{P_{excess} = \frac{2S}{r}} \qquad \boxed{P_{excess} = \frac{4S}{r}}$$

→ If two drops/ bubbles are in contact, then to get the curvature of common surface,

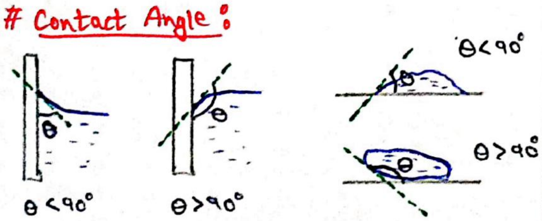
$$\boxed{P \propto \frac{1}{R}}$$ → Smaller one exert more pressure



$$\boxed{r_3 = \frac{r_1 r_2}{r_2 - r_1}} \qquad \boxed{r_3 = \frac{r_1 r_2}{r_1 + r_2}}$$

[r_3 → Radius of curvature of common surface]

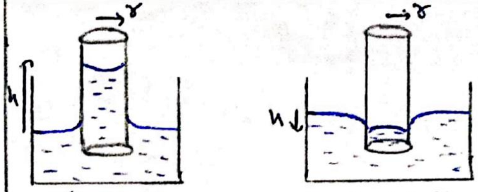
Contact Angle :



In case of (θ>90°), cohesive forces among fluid particles dominate ($F_C > F_A$)

Cappillary Action :

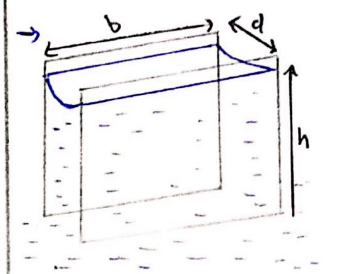
→ Depending upon fluid, two types of meniscus can be observed in a cappillary.



$$\boxed{h = \frac{2S\cos\theta}{r\rho g}}$$

r → Radius of meniscus

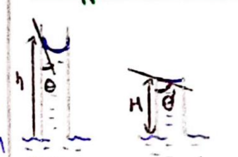
For water, θ=0°



$$\boxed{h = \frac{2S\cos\theta}{d\rho g}}$$

For water, θ=0°

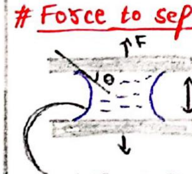
Insufficient tube length :→ (H<h)



In such case, liquid won't overflow. The meniscus adjust itself (↑ contact angle)

$$\boxed{\theta' > \theta}$$

Force to separate Glass plate :



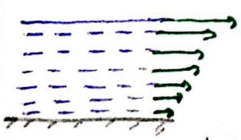
$$\boxed{F = \frac{2SA}{d}\cos\theta}$$

(For water, θ=0°)

A : Liq. cross section

Viscous Force :

→ Fluid opposes the relative motion of its different layers.



$$\boxed{F_v = -\eta A\frac{dv}{dy}}$$

A : Cross section of fluid

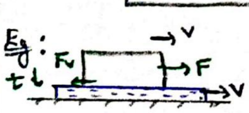
$[\eta] = M\cdot L^{-1}\cdot T^{-1}$ | $\frac{dv}{dy}$: Velocity gradient (vertically)

SI Unit : Poiseuille (Pl)

CGS Unit : Poise

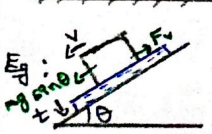
$$\boxed{1\,Pl = 10\text{ Poise}}$$

Eg :



$F = F_v = \eta A\frac{v}{t}$

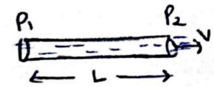
Eg :



$mg\sin\theta = \eta A\frac{v}{t}$

Poiseuille's Eqⁿ :

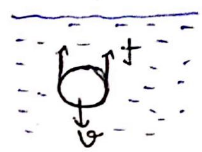
→ When a viscous fluid flows through a narrow tube, pressure energy /vol. is lost due to its viscosity.



Flow rate.
$$\boxed{Q = \frac{\pi r^4}{8\eta L}(P_1 - P_2)}$$

Stoke's law & Terminal Velocity :

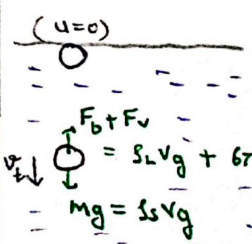
→ If a spherical ~~velocity~~ body is moving downward in a fluid, then fluid applies viscous force in opp dirⁿ along with buoyant force.



$$\boxed{f = 6\pi\eta r v}$$

(Total force) of viscosity

→ The instant at which buoyant force & viscous force becomes equal to body's weight, its accⁿ becomes zero & velocity becomes constant.
↳(Terminal velocity)



$F_b + F_v = \rho_L V g + 6\pi\eta r v$

$Mg = \rho_s V g$

$$\boxed{v_t = \frac{2r^2 g}{9\eta}(\rho_s - \rho_L)}$$